CocoaSense AI — forecast report

Generated 05 November 2026 · data through October 2026

US$5,750/t

down 2.9% on the month

October 2026 monthly average · about GH¢66,873/t at GH¢11.63 per US$

Forecasts for the coming months

Month	Ahead	Model	US$/t	Change	GH¢/t	Confidence
Nov 2026	1 mo	Consensus	5,677	-1.3%	66,019	
		ARIMA	5,814	+1.1%	67,621	Low
		Naive	5,750	+0.0%	66,873	Low
		Random Forest	5,679	-1.2%	66,047	Low
		XGBoost	5,537	-3.7%	64,391	Low
Nov 2026	3 mo	Consensus	5,800	+0.9%	67,453	
		ARIMA	5,971	+0.4%	69,442	Low
		Naive	5,950	+0.0%	69,199	Low
		Random Forest	5,916	-0.6%	68,798	Low
		XGBoost	5,513	-7.3%	64,120	Low
Dec 2026	3 mo	Consensus	5,706	-0.8%	66,360	
		ARIMA	5,841	-1.3%	67,931	Low
		Naive	5,920	+0.0%	68,850	Low
		Random Forest	5,842	-1.3%	67,947	Low
		XGBoost	5,434	-8.2%	63,201	Low
Jan 2027	3 mo	Consensus	5,611	-2.4%	65,251	
		ARIMA	5,791	+0.7%	67,345	Low
		Naive	5,750	+0.0%	66,873	Low
		Random Forest	5,662	-1.5%	65,854	Low
		XGBoost	5,379	-6.5%	62,552	Low

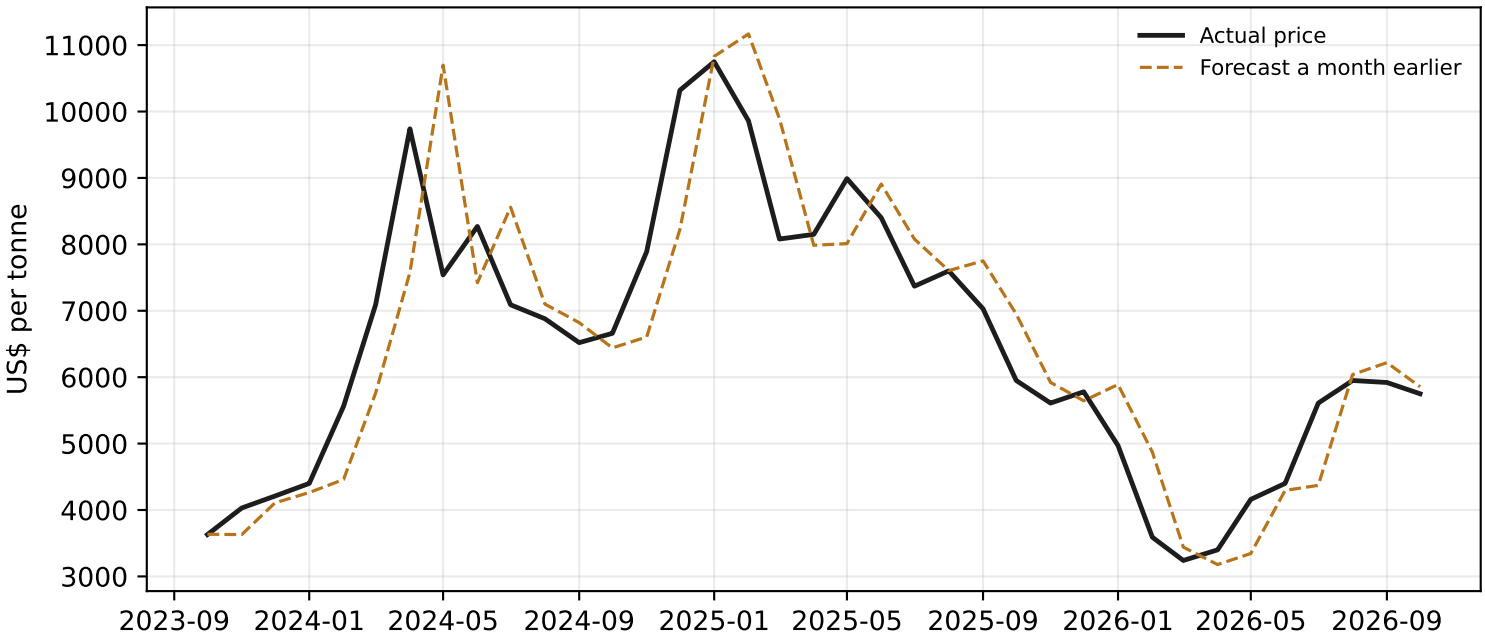
Confidence comes from how close each model's earlier forecasts were, adjusted for how volatile the market is now. Low confidence means the actual price could easily land 10% or more away from the forecast. Cedi values use today's exchange rate, because future exchange rates are not forecast.

Simulated results on public prices. These are not Ghana's or COCOBOD's actual gains or losses, and the forecasts are not trading advice. Frozen models were trained only on data up to December 2019; learning models are refitted each month on prices known at the time.

Track record

How often past forecasts were right (2020 onwards)

Last three years: actual price and the consensus forecast



Success and failure rates

Models	Ahead	Model	Forecasts	Within ±10%	Missed	Within ±5%
Frozen	1 mo	ARIMA	82	79%	21%	61%
Frozen	1 mo	LSTM	82	71%	29%	54%
Frozen	1 mo	Naive	82	79%	21%	59%
Frozen	1 mo	Random Forest	82	77%	23%	62%
Frozen	1 mo	XGBoost	82	76%	24%	55%
Frozen	3 mo	ARIMA	82	51%	49%	28%
Frozen	3 mo	LSTM	82	45%	55%	22%
Frozen	3 mo	Naive	82	52%	48%	28%
Frozen	3 mo	Random Forest	82	50%	50%	24%
Frozen	3 mo	XGBoost	82	49%	51%	27%
Learning	1 mo	ARIMA	82	76%	24%	62%
Learning	1 mo	Naive	82	79%	21%	59%
Learning	1 mo	Random Forest	82	78%	22%	61%
Learning	1 mo	XGBoost	82	78%	22%	54%
Learning	3 mo	ARIMA	82	50%	50%	28%
Learning	3 mo	Naive	82	52%	48%	28%
Learning	3 mo	Random Forest	82	48%	52%	26%
Learning	3 mo	XGBoost	82	45%	55%	24%

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Key findings

Held-out evaluation, January 2020 to the latest month

- No model reliably beat the naive 'price stays the same' forecast. ARIMA had the lowest error (6.87% at 1 month, 14.88% at 3 months), only 3.6% and 1.1% better than naive and not statistically significant.
- One month ahead, about 3 forecasts in 4 landed within ±10% of the actual price; three months ahead, about half. In calm 2020–2022 about 9 in 10 did.
- The 2023–25 surge broke the models: they had learned from 1991–2019 that high prices fall back, monthly volatility doubled (about 6.5% to 13.5%), and the causes (weather, disease, smuggling) are not in price data.
- The forecast-driven selling rule earned 0.8% less than a fixed 50/50 split. A volatility guard narrowed the gap to 0.4% but did not close it.
- Volatility-aware prediction ranges brought XGBoost's 80% range coverage from 72% to 84% (1 month): uncertainty can be measured honestly even when the price cannot be predicted.
- The cedi exchange rate did not improve forecasts. Refitting the models every month did not improve accuracy either.
- Conclusion: on public price data, machine learning does not improve forward-sales revenue. Its value lies in measuring uncertainty, which tells a seller when a fixed selling split is the safer choice.

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